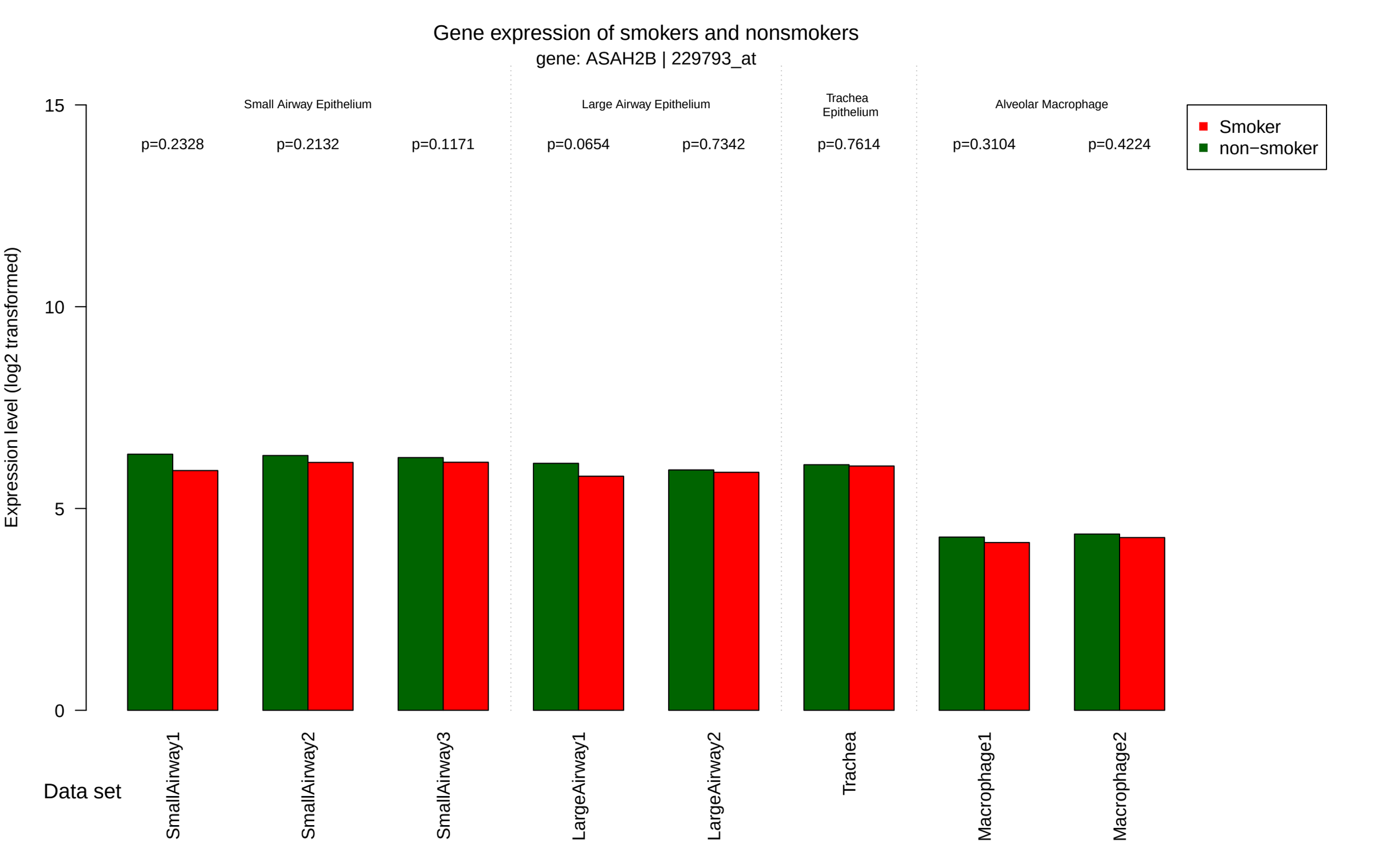
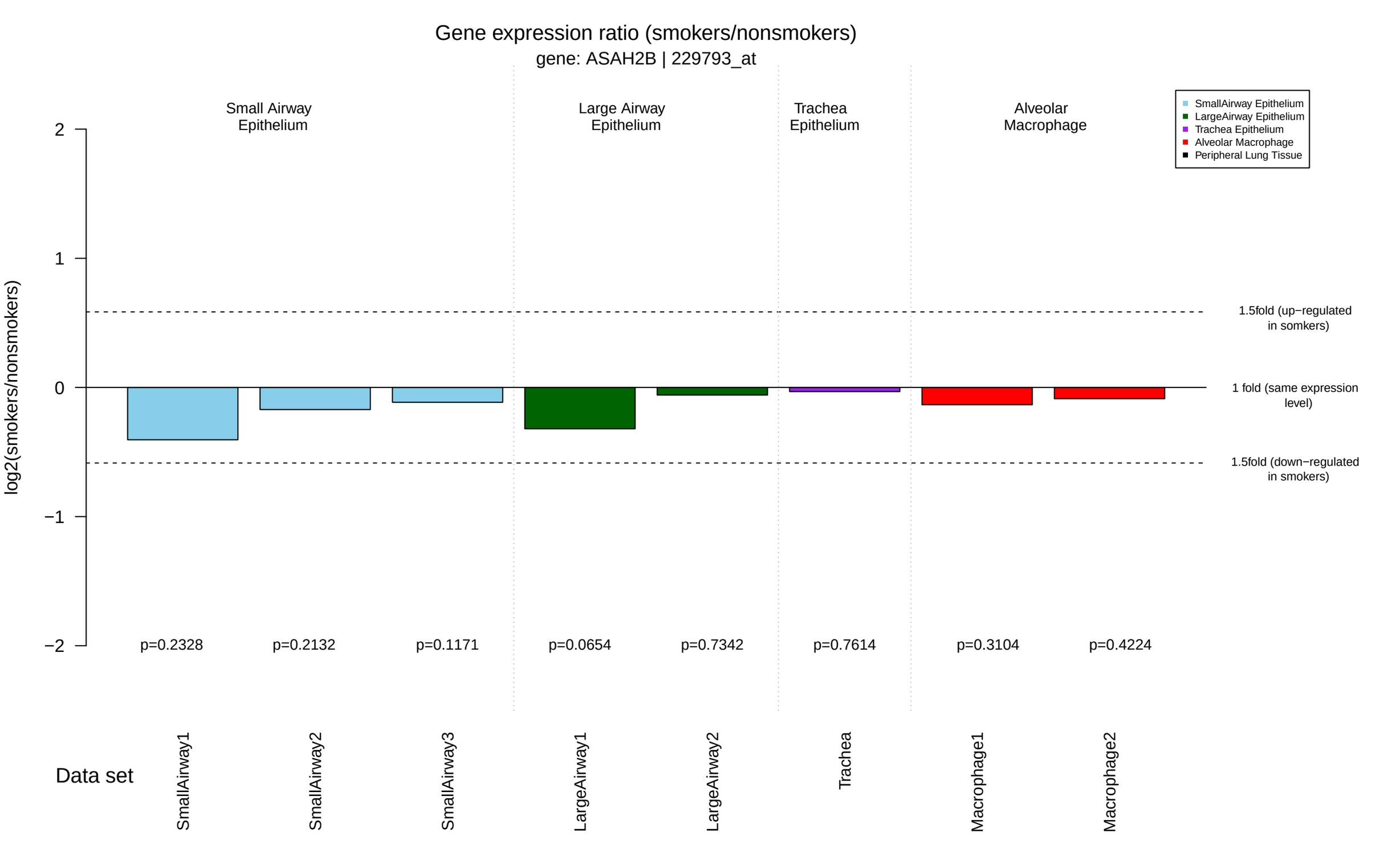
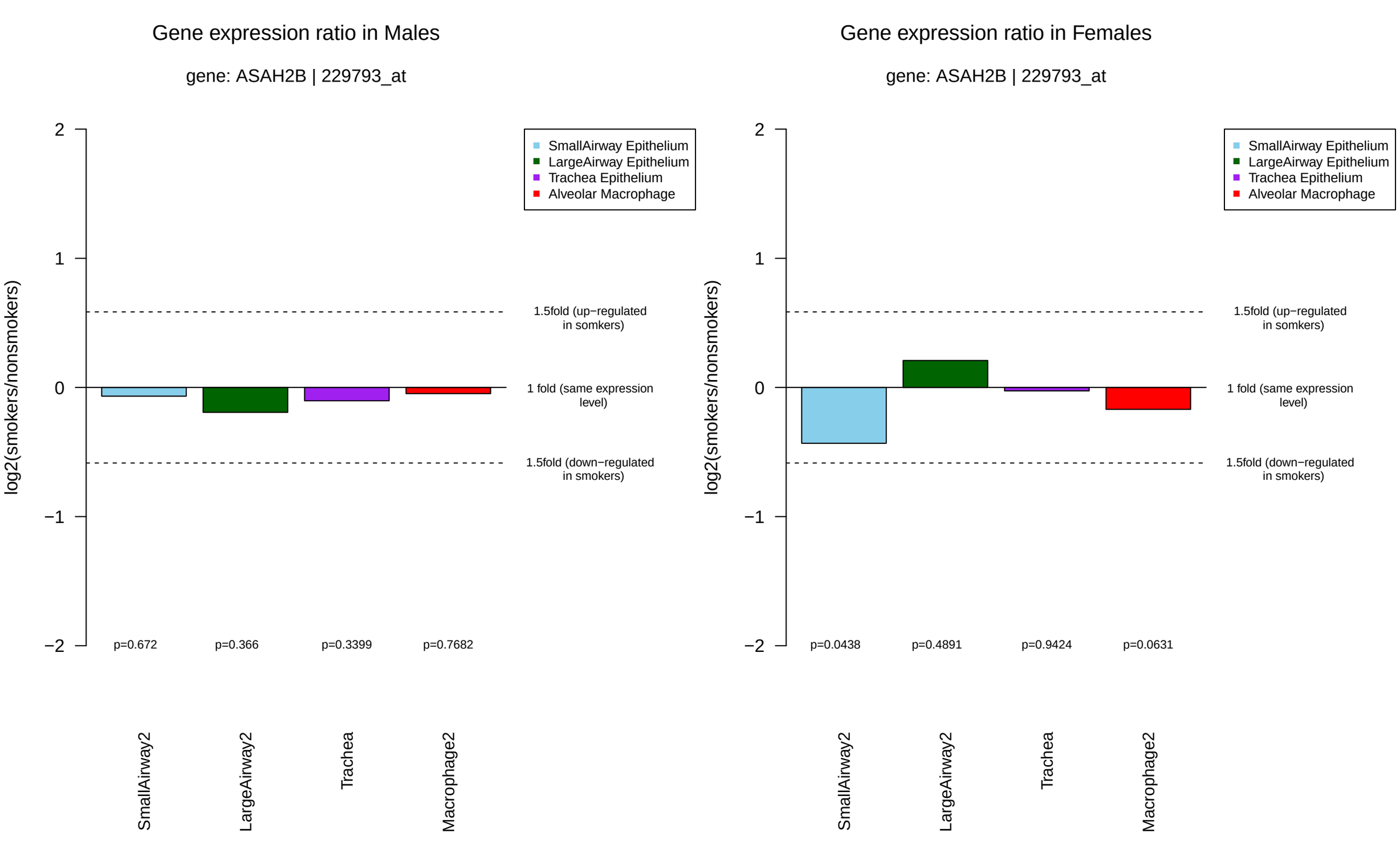
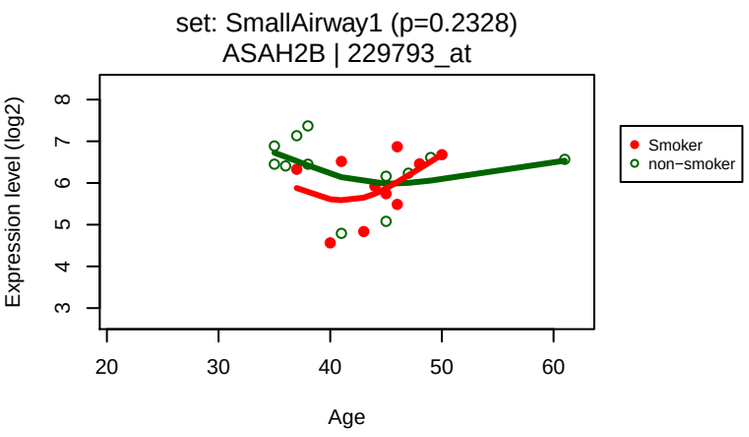
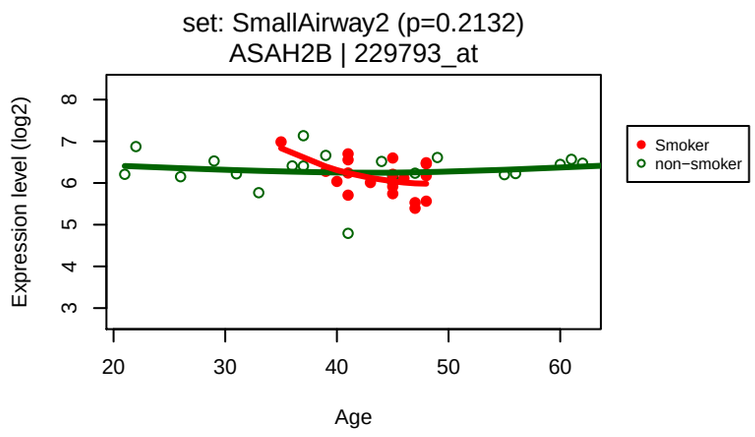
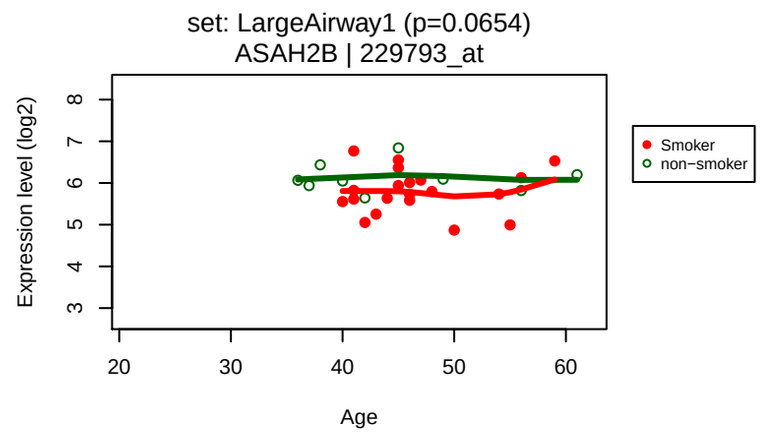

Gene expression of smokers and nonsmokers
gene: ASAH2B | 229793_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2328
p=0.2132
p=0.1171
p=0.0654
p=0.7342
p=0.7614
p=0.3104
p=0.4224
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

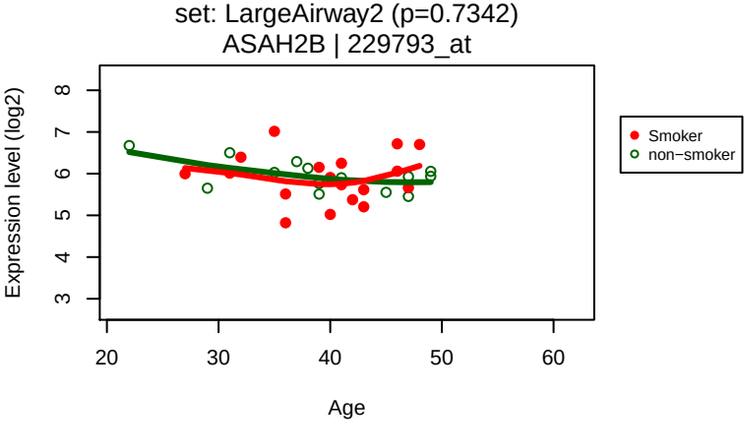

Gene expression ratio (smokers/nonsmokers)
gene: ASAH2B | 229793_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2328
p=0.2132
p=0.1171
p=0.0654
p=0.7342
p=0.7614
p=0.3104
p=0.4224
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

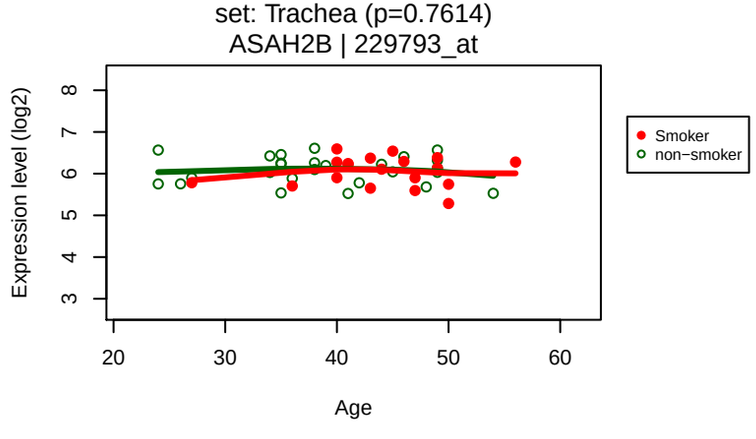

Gene expression ratio in Males
gene: ASAH2B | 229793_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.672
p=0.366
p=0.3399
p=0.7682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASAH2B | 229793_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0438
p=0.4891
p=0.9424
p=0.0631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

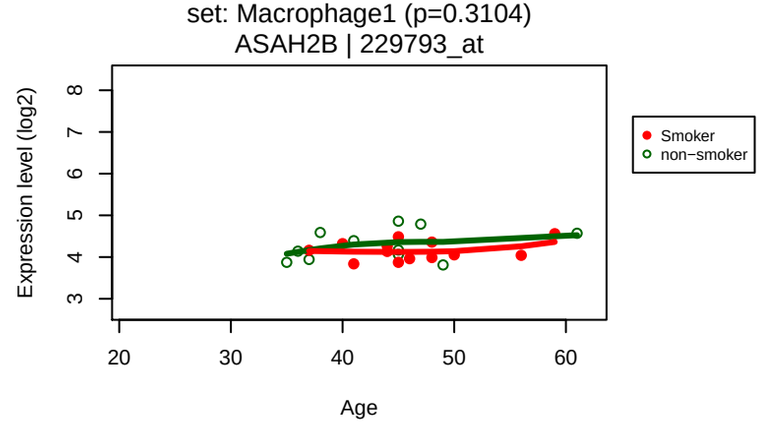

set: SmallAirway1 (p=0.2328)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker

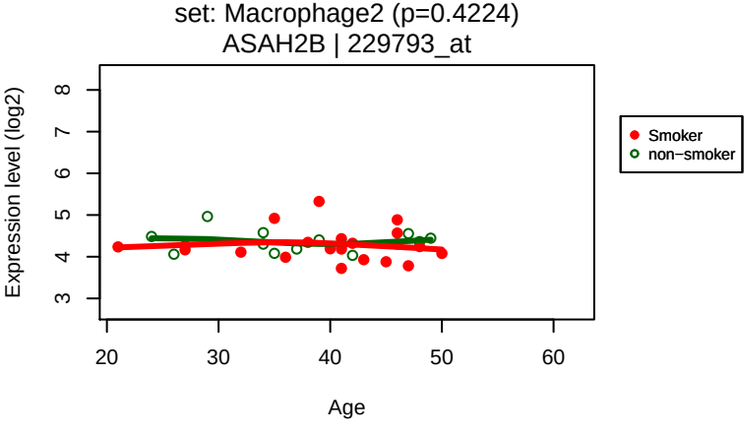

set: SmallAirway2 (p=0.2132)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker

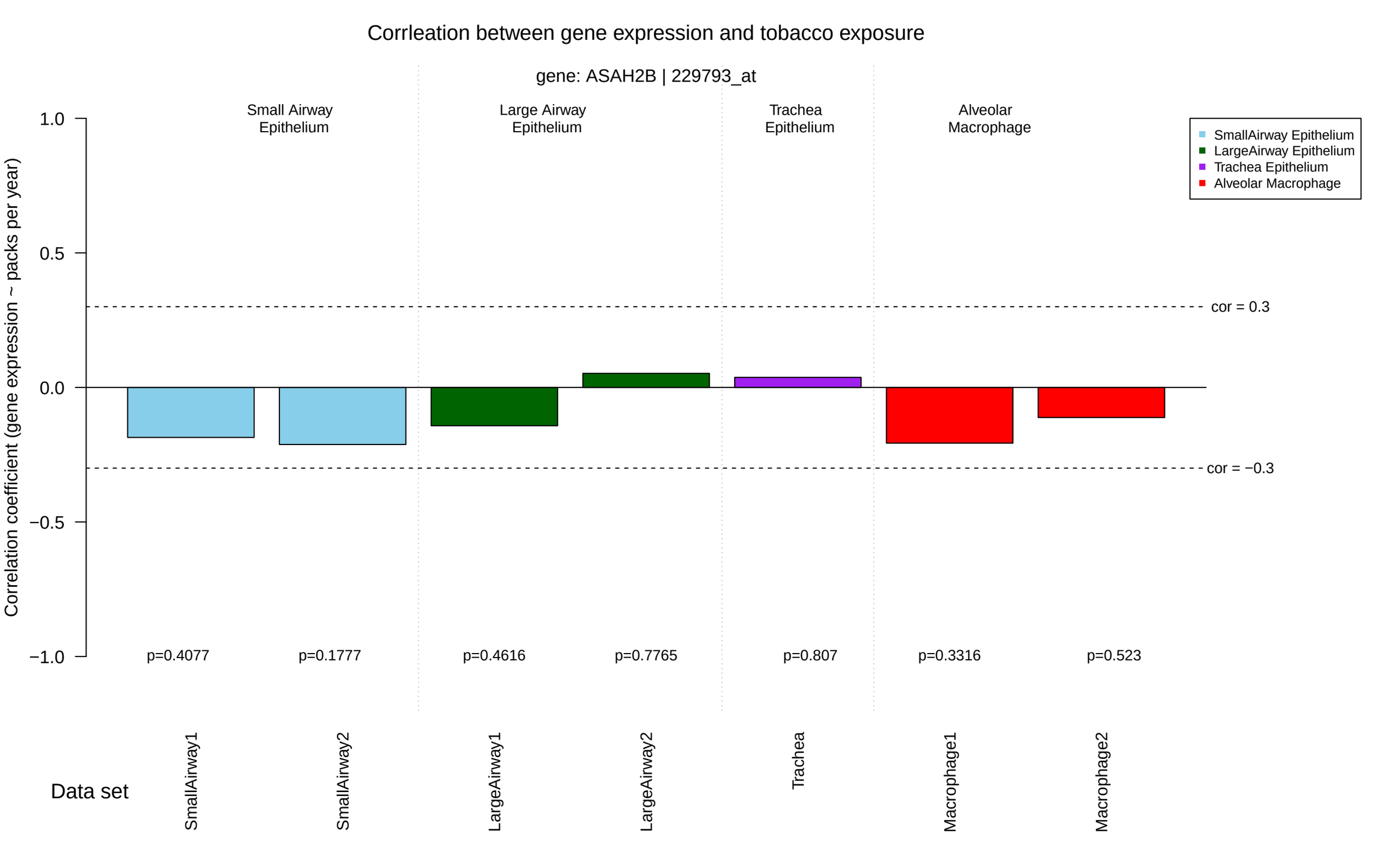

set: LargeAirway1 (p=0.0654)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7342)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7614)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3104)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4224)
ASAH2B | 229793_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASAH2B | 229793_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4077
p=0.1777
p=0.4616
p=0.7765
p=0.807
p=0.3316
p=0.523
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2